



Institute of Computer Engineering Technology



iCET Certified Master

ASSIGNMENT

Module	Database Management System
Assignment Title	Data Definition Language - DDL
Assignment Type	Practical Exercise

01) Create the following database and submit SQL commands for the given task.

This is regarding an airport management system, along with some sample data, with the following tables.

- Database name: **AirportManagement**
- Tables: Airline, Flight, Ticket, Passenger

```
mysql> DESC Airline;
```

Field	Type	Null	Key	Default	Extra
airline_id	int	YES		NULL	
airline_name	varchar(100)	YES		NULL	
country	varchar(50)	YES		NULL	

```
mysql> DESC Flight;
```

Field	Type	Null	Key	Default	Extra
flight_id	int	YES		NULL	
airline_id	int	YES		NULL	
flight_number	varchar(20)	YES		NULL	
origin	varchar(50)	YES		NULL	
destination	varchar(50)	YES		NULL	
departure_time	datetime	YES		NULL	
arrival_time	datetime	YES		NULL	

```
mysql> DESC Ticket;
```

Field	Type	Null	Key	Default	Extra
ticket_id	int	YES		NULL	
flight_id	int	YES		NULL	
ticket_price	decimal(10,2)	YES		NULL	
ticket_class	varchar(20)	YES		NULL	

```
mysql> DESC Passenger;
```

Field	Type	Null	Key	Default	Extra
passenger_id	int	YES		NULL	
first_name	varchar(50)	YES		NULL	
last_name	varchar(50)	YES		NULL	
email	varchar(100)	YES		NULL	
age	int	YES		NULL	
seat_number	varchar(10)	YES		NULL	
ticket_id	int	YES		NULL	

- Tables with sample records :

```
mysql> SELECT * FROM Airline;
```

airline_id	airline_name	country
1	SkyWings	USA
2	BlueBird Airways	UK
3	Oceanic Air	Australia
4	FalconFly	UAE
5	GlobalJet	Germany

```
mysql> SELECT * FROM Flight;
```

flight_id	airline_id	flight_number	origin	destination	departure_time	arrival_time
1	1	SW101	New York	Los Angeles	2025-08-01 08:00:00	2025-08-01 11:30:00
2	2	BB202	London	Paris	2025-08-02 09:00:00	2025-08-02 10:20:00
3	3	OA303	Sydney	Melbourne	2025-08-03 07:15:00	2025-08-03 08:45:00
4	4	FF404	Dubai	Mumbai	2025-08-04 14:00:00	2025-08-04 17:15:00
5	5	GJ505	Berlin	Rome	2025-08-05 10:00:00	2025-08-05 12:10:00

```
mysql> SELECT * FROM Ticket;
```

ticket_id	flight_id	ticket_price	ticket_class
1	1	199.99	Economy
2	2	89.50	Business
3	3	120.75	Economy
4	4	220.00	First
5	5	150.00	Economy

```
mysql> SELECT * FROM Passenger;
```

passenger_id	first_name	last_name	email	age	seat_number	ticket_id
1	John	Doe	john.doe@example.com	30	12A	1
2	Emily	Smith	emily.smith@example.com	27	4B	2
3	Carlos	Garcia	carlos.garcia@example.com	35	8C	3
4	Aisha	Khan	aisha.khan@example.com	22	1A	4
5	Luca	Rossi	luca.rossi@example.com	40	16D	5

Answer the following questions based on the given airline database.

01)

1. Add a new column called **rating** of type **DECIMAL(3,1)** to the **Airline** table to store customer ratings
2. Rename the column **seat_number** in the **Passenger** table to **seat_code**.
3. Modify the **ticket_class** column in the **Ticket** table to increase its length to **VARCHAR(50)**.
4. Drop the column **age** from the **Passenger** table.
5. Add a new column named **status** of type **VARCHAR(20)** to the **Flight** table to store flight statuses such as "On Time", "Delayed", or "Cancelled".

02) Design a database on the information below

- Database name: **Bookshop**
- Tables: Book, Author, Customer, Order, Review

Tables:

- **Book**
 - ISBN
 - Name
 - Price
 - Publish Date
- **Author**
 - AuthorID
 - Name
 - Biography
- **Customer**
 - CustomerID
 - Name
 - Email
 - Address
- **Orders**
 - OrderID
 - CustomerID
 - Order Date
- **Review**
 - ReviewID
 - Rating
 - Comment

Answer the following questions based on the Bookshop database given.

1. Create the tables and insert the sample 8 records
 - a. Add a new column called **Genre** of type **VARCHAR(50)** to the **Book** table.
 - b. Add a new column called **Phone** of type **VARCHAR(15)** to the **Customer** table.
 - c. Rename the column **OrderDate** in the **Orders** table to **PurchaseDate**.

03) Create a new database for a small shop and create tables for orders and customers. Write complete SQL codes for the given task.

- Design SQL statements to create tables for customers and orders.
 - The name of the shop is **puffycave**
 - The Customers and Order table contains information about visitors to the cafe.
 - Customer fields - **customer_id, first_name, last_name, email, phone, address, city, created_at**
 - Orders fields - **order_id, customer_id, order_date, order_total, payment_method, order_status, notes**
- Create the tables and insert sample 10 records

04) Create a database named "LCport" and use it.

4.1) Create a table named "Ships" with columns for ship id (integer), ship name (varchar), origin port (varchar), destination port (varchar), arrival date (DATE) and departure date (DATE) and insert sample 5 records.

4.2) Create a table named "Containers" with columns for container id (integer), ship id (Integer), container type (varchar), weight (decimal) and status (varchar) and insert sample 5 records.

4.3) Create a table named "Cargo" with columns for cargo id (integer), container id (Integer), description (varchar), weight (decimal) and cargo type (varchar) and insert sample 5 records.

4.4) Create a table named "Employees" with columns for employee id (integer), full name (varchar), role (varchar), shift (varchar) and contact number (varchar) and insert sample 5 records.

4.5) Create a table named "Schedules" with columns for schedule id (integer), ship id (integer), berth number (varchar), scheduled arrival (DATETIME) and scheduled departure (DATETIME) and insert sample 5 records.

How to Submit the Assignment

01) Provide a screenshot displaying the structure of the specified tables.

02) Insert sample records into the tables and provide screenshots showing all details.